

User-Adaptive Verbalization of Explanations for Explainable AI

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Student Declaration

I hereby declare that the work presented in the report entitled ”**User-Adaptive Verbalization of Explanations for Explainable AI**” submitted by me for the partial fulfilment of the requirements for the degree of *B.Tech. in Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under the guidance of **Dr. Raghava Mutharaju** . Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

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Place & Date: May 2, 2026

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dr. Raghava Mutharaju

Place & Date: May 2, 2026

Abstract

This work explores an ontology-grounded framework for generating user-adaptive explanations across multiple biomedical tasks. To achieve this, we combine a biomedical language model, local feature explanations, and biomedical ontologies to produce readable justifications tailored to different levels of domain expertise. In addition, the project investigates how steering vectors could be used to further control the style and complexity of these explanations, outlining this as a direction for future development.

Keywords: (Explainable AI, LIME, ontology-based explanations, knowledge graphs, user-adaptive explanations, beginner and expert users, steering vectors, clinical language models, natural language justifications, biomedical tasks.)

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Chapter 1

Introduction

1.1 Background

Deep learning has demonstrated remarkable performance across biomedical tasks such as disease classification, diagnostic assistance, and phenotyping. However, these models often function as opaque “black boxes,” making it difficult to understand the reasoning behind their decisions. This opacity poses substantial risks in biomedical settings, where reliability, accountability, and trust are essential [12]. As highlighted by recent surveys, a major barrier to deploying machine learning in healthcare is the lack of transparent and trustworthy explanations that clinicians and patients can meaningfully interpret [1].

The rise of Explainable Artificial Intelligence (XAI) attempts to address these challenges by offering post-hoc interpretability methods. Prominent techniques such as LIME [12] and SHAP [7] attribute feature importance to model predictions. While these approaches provide valuable insights into model behaviour, research has consistently shown that they suffer from critical limitations: they generate explanations that are local but not globally consistent, lack semantic grounding, are sensitive to perturbations, and often fail to align with human cognitive expectations [8]. Furthermore, most current XAI methods focus exclusively on numerical or visual artefacts rather than producing coherent, domain-specific natural language explanations [1].

Recent research converges on the idea that combining symbolic Knowledge Representation and Reasoning (KRR) techniques with data-driven Machine Learning (ML) can yield hybrid intelligent systems with improved interpretability [3]. Symbolic structures such as ontologies encode hierarchies, biomedical semantics, and clinically validated conceptual relationships, enabling interpretability that is more aligned with human reasoning. As emphasized in [8], trustable XAI must go beyond surface-level feature importance and incorporate domain-specific knowledge to generate explanations that are verifiable, contextually meaningful, and actionable.

A second major challenge is that explanations must be *user-adaptive*. According to recent findings in adaptive XAI [13], the effectiveness of an explanation depends heavily on the user’s expertise, cognitive load, and familiarity with domain concepts. A novice user often benefits from simpli-

fied, high-level descriptions, whereas experts require deeper, mechanistic explanations referencing biomedical ontology terms. Traditional XAI techniques, however, produce static and homogeneous explanations regardless of user background, limiting their practical utility.

Taken together, these insights highlight two significant gaps in biomedical XAI: (i) the absence of ontology-grounded semantic explanations, and (ii) the lack of user-adaptive natural language explanations. Bridging these gaps requires integrating XAI, biomedical ontological reasoning, and modern generative language models into a unified pipeline capable of producing meaningful, context-aware narratives tailored to the user.

A further dimension of this challenge concerns *generation-time control*. Even when an appropriate LLM is conditioned on the correct prompt and ontological context, unconstrained decoding can produce explanations that are lexically too complex for lay users. Two complementary inference-time control strategies have recently emerged: *constrained decoding*, which modifies token-level probability distributions at each generation step to penalise linguistically complex vocabulary [10], and *steering vectors*, which manipulate intermediate model representations to shape the stylistic register of the output. In this work, we study both of these mechanisms as candidate strategies for enforcing user-appropriate readability in biomedical explanations without requiring model retraining.

1.2 Motivation

Biomedical datasets inherently contain domain semantics, yet classical XAI methods do not map feature-level explanations to clinically validated concepts. Ontologies such as UMLS, SNOMED-CT, and MeSH provide structured representations of biomedical knowledge, including hierarchical relations, synonyms, semantic types, and clinically meaningful groupings [4]. Leveraging these ontologies enables the transformation of raw feature importances into explanations that align with biomedical reasoning processes.

In parallel, modern large language models (LLMs) such as Qwen2.5, BioBART, and GPT have demonstrated powerful abilities in natural language generation, contextual integration, and explanation synthesis. However, without explicit biomedical context, LLM-generated explanations can suffer from hallucination, lack of factual grounding, or oversimplification. The literature emphasizes that explainability must connect model behaviour to domain knowledge in a way that is both accurate and comprehensible [1, 8].

Furthermore, as highlighted in recent adaptive-explanation frameworks [13], explanations should be personalized to the recipient’s level of expertise. A one-size-fits-all explanation risks overwhelming novice users or under-informing clinical experts. Therefore, the integration of ontological reasoning with user profiling and LLM-based generation constitutes a promising approach to producing explanations that are both semantically meaningful and adaptively tailored.

This work focuses specifically on the task of *multi-class biomedical abstract classification*, a clinically relevant setting in which a model assigns a medical text to one of several disease category groups. This constrained task provides a well-defined prediction surface against which the explainability

pipeline can be rigorously evaluated.

Thus, combining (i) feature-level XAI techniques (LIME and Integrated Gradients), (ii) ontology-driven semantic enrichment, (iii) user-adaptive transformer-based natural language generation, and (iv) inference-time control mechanisms offers a path toward practical, trustworthy, and clinically relevant explainability in biomedical models. Within this design space, this work investigates two complementary approaches to controlling explanation difficulty: an ontology-grounded pipeline with readability-constrained decoding, and a steering-vector-based approach that modulates generation style through hidden-state intervention. Studying both approaches enables a comparison between explicit symbolic grounding and latent activation-based control for user-adaptive biomedical explanation generation.

1.3 Problem Definition

Biomedical machine learning models produce high-performing classification outputs, yet their decision-making processes remain opaque and insufficiently interpretable. In the specific setting considered in this work — multi-class classification of biomedical abstracts into disease categories — a fine-tuned classifier assigns each abstract to one of five groups (Neoplasms, Digestive system diseases, Nervous system diseases, Cardiovascular diseases, or General pathological conditions). Given an input sample x and its corresponding model prediction $f(x)$, existing techniques such as LIME [12] and Integrated Gradients [15] identify influential features but provide no structured biomedical meaning, no ontological contextualization, and no coherent natural-language explanation.

Moreover, these explanations remain static and undifferentiated across users regardless of their expertise, violating the principles of adaptive and trustable explanation design described in recent literature [13, 1]. In clinical scenarios, explanatory needs vary widely across medical professionals, researchers, and lay users, making fixed explanations impractical and potentially misleading. Standard prompt-based audience conditioning of LLMs is additionally insufficient: without explicit generation-time constraints, the model may produce vocabulary well beyond the comprehension of a lay reader regardless of the instruction provided.

Thus, the central problem addressed in this work is:

How can raw feature-based XAI outputs be transformed into ontology-grounded, context-aware, and user-specific natural language explanations — with verifiable readability characteristics — suitable for different expertise levels in biomedical multi-class classification, and how do explicit decoding-based control and latent steering-based control compare as mechanisms for achieving this adaptation?

Answering this requires a hybrid approach that integrates XAI, ontological reasoning, user modeling, and inference-time controlled natural language generation.

1.4 Proposed Approach

To address the problem defined above, this work investigates two complementary pipelines for user-adaptive biomedical explanation generation. The first combines classifier-based XAI, ontology-grounded semantic enrichment, and readability-constrained decoding. The second is a steering-vector-based pipeline that controls explanation difficulty by intervening directly in the hidden-state representations of the language model during generation. Together, these pipelines enable a comparison between explicit symbolic grounding and latent activation-based control as mechanisms for producing **beginner**, **intermediate** and **expert-level** explanations.

Pipeline A: Ontology-Grounded XAI with Constrained Decoding

The first pipeline operates on the task of multi-class biomedical abstract classification and produces explanations that are grounded in both model evidence and biomedical ontology structure. Its stages are as follows:

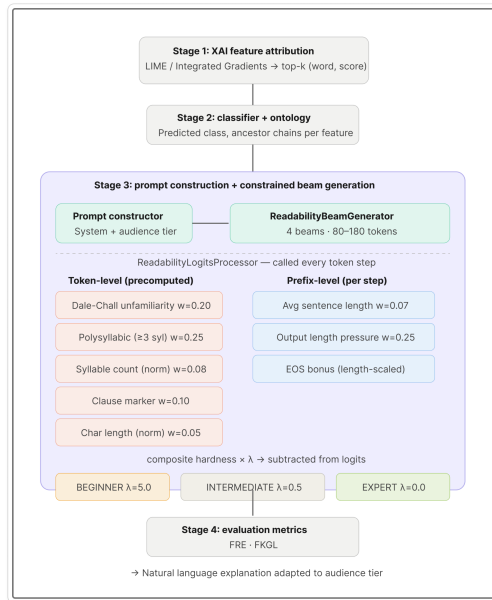


Figure 1.1: Overview of the readability-constrained beam decoding pipeline.

- **Task model prediction:** A fine-tuned biomedical classifier assigns each abstract to one of five disease-related categories and outputs a confidence score.
- **Feature importance extraction:** A pluggable XAI module identifies the most influential input tokens. The pipeline supports both LIME [12], which estimates local feature importance through perturbation, and Integrated Gradients [15], which computes gradient-based token attributions at the embedding level.

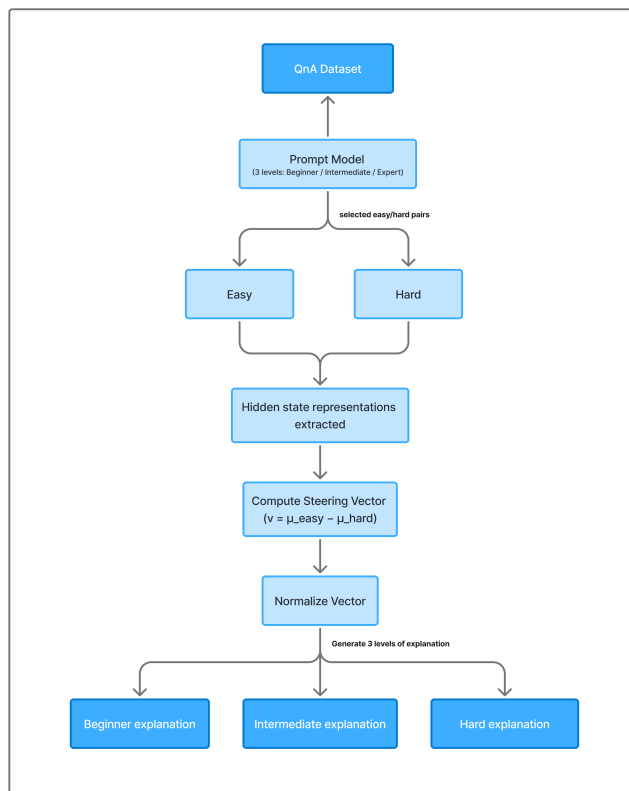


Figure 1.2: Steering Vector Pipeline

- **Ontology-based semantic enrichment:** The most salient features are mapped to biomedical ontologies such as DOID and MeSH. For each resolved concept, the system retrieves ontology ancestors and semantic context. User-appropriate ancestor chains are then selected using ontology-derived difficulty signals such as specificity and popularity.
- **User-adaptive explanation generation:** The predicted class, attributed features, ontology context, and user category are combined into a structured prompt for the language model.
- **Readability-constrained decoding:** Explanation generation is performed using a custom constrained beam-search strategy. A **LogitsProcessor** penalises lexically complex candidate tokens at inference time using weighted readability heuristics, including Dale-Chall familiarity, syllable count, polysyllabic rate, clause density, and character length. The penalty strength is adjusted according to audience level, enabling explicit control over output difficulty at the token level.

Pipeline B: Steering-Vector-Based Difficulty Control

The second pipeline explores a latent control mechanism for explanation difficulty. Instead of explicitly penalising difficult words during decoding, it modifies the internal hidden-state dynamics

of the language model using steering vectors. Its stages are as follows:

- **Contrastive pair construction:** Easier and harder explanation pairs are collected using prompts or generated outputs representing different intended difficulty levels.
- **Steering vector extraction:** Hidden-state representations are extracted from an intermediate layer of the language model for these contrastive examples. A steering direction is computed as the difference between the average representations of easy and hard explanations.
- **Activation-space intervention:** During generation, the steering vector is injected into a selected model layer using a tunable scaling coefficient. This modifies the style and complexity of the explanation without changing the model weights.
- **Difficulty control through scaling:** By varying the steering coefficient, the system can continuously shift the output toward simpler or more technical language.
- **Biomedical evaluation:** The steered model is evaluated on biomedical text in order to assess whether latent activation manipulation can generate user-adaptive explanations comparable to the ontology-grounded constrained decoding approach.

In this work, we empirically compare the outputs of both pipelines across readability metrics (FRE and FKGL), semantic grounding, and adaptability to user expertise levels. This comparison allows us to analyse the trade-offs between explicit, ontology-driven control and latent, representation-level steering for biomedical explanation generation.

Chapter 2

Literature Review

This chapter reviews existing works on healthcare knowledge graphs, ontology engineering, difficulty modelling, biomedical language models, explainable AI, steering vectors, and user-adaptive XAI, and positions the proposed framework within this landscape.

2.1 Knowledge Graphs in Biomedical Domain

Healthcare knowledge graphs provide the structural backbone for this project. Yang et al. survey HKGs as semantic graphs that capture domain entities such as diseases, drugs, genes, and phenotypes along with their relationships, and describe how they are built either from scratch or by integrating existing ontologies and clinical data sources. Their review distinguishes model-free uses (e.g., querying, visualization) from model-based uses in which HKGs enrich downstream machine learning models, concluding that HKGs are key enablers for clinical decision support and interpretable analytics. This motivates the choice to ground explanations in disease and symptom knowledge graphs rather than relying only on raw text features [20].

While Yang et al. argue for the importance of HKGs in general, Azim and Imran provide a practical perspective on how biomedical knowledge graphs are actually constructed. They outline a pipeline starting from curated databases and literature, moving through ontology-based data access and integration, and using NLP techniques such as named entity recognition and relation extraction to populate graphs. Their case studies demonstrate that well-designed biomedical KGs can integrate heterogeneous data and support tasks ranging from information retrieval to hypothesis generation, which aligns directly with this project’s use of existing disease and symptom ontologies as an external source of medically grounded structure for explanation [2].

2.2 Ontology Engineering and Difficulty Modelling

Both of these works assume robust ontologies underlie the graphs, which leads naturally to ontology engineering guidelines. Noy and McGuinness describe practical steps for building ontologies,

including identifying key domain terms, deciding what should be modeled as a class versus an instance, designing class hierarchies, and defining properties and constraints. They stress choosing an appropriate level of granularity for the intended application and highlight how OWL reasoners can classify ontologies, infer new subclass relationships, and ensure logical consistency. These principles inform how disease and symptom ontologies are selected and used in this project, and why reasoners are employed to pull ancestors, related conditions, and definitions when constructing explanations [9].

Beyond representing knowledge, this work needs a way to relate ontological structure to user expertise. Vinu and Kumar propose a difficulty-level model for ontology-based factual questions that combines ontology-derived features with Item Response Theory. They quantify question difficulty using metrics such as popularity, specificity, selectivity, and coherence, and train separate logistic regression models for experts, intermediates, and beginners to predict difficulty labels. Their results show that this feature set predicts perceived difficulty more effectively than earlier methods, which directly inspires the approach in this project: ontology concepts and ancestors are scored using similar structural features and grouped into beginner-friendly or expert-level context [18].

2.3 Biomedical Language Models and Tasks

Complementing the symbolic layer, modern biomedical NLP models provide the base predictions that must be explained. Lee et al. introduce BioBERT, a domain-specific adaptation of BERT further pre-trained on large PubMed and PMC corpora to better capture biomedical terminology and syntax. Their experiments demonstrate consistent performance gains over general-domain BERT on several biomedical benchmarks, confirming that in-domain pretraining is crucial for biomedical tasks. This supports the use of biomedical transformers like BioBERT as the underlying prediction models in this project [6].

2.4 Explainable AI and Ontology-Grounded Explanations

Explainable AI methods link model predictions with interpretable features that can then be grounded in an ontology. Ribeiro, Singh, and Guestrin present LIME, a model-agnostic framework that fits local surrogate models around individual predictions to highlight the most influential features. Their case studies show how LIME helps detect data leakage, uncover spurious correlations, and improve transparency in text classification. In this project, LIME extracts key biomedical tokens and phrases from each prediction, which are subsequently mapped onto ontology entities and expanded using reasoner-derived relations [12].

2.5 Controllable and User-Adaptive Explanations

Recent work on controllable generation shows that language model outputs can be shaped at inference time without retraining. Subramani et al. demonstrate that pretrained models contain latent steering vectors—directions in activation space that can modify semantic and stylistic attributes when injected into intermediate layers [14]. Building on this idea, more recent approaches focus on instruction-level control through attention manipulation. In particular, venkateswaran-contractor-2026-spotlight propose *SpotLight*, which dynamically reallocates attention toward user-specified instruction spans by adjusting attention proportions during generation, enabling stable and adaptive control without over-steering.

A key contribution of this work is the evaluation of instruction-following using benchmarks such as *IFEval*, which assesses adherence to multiple syntactic constraints through prompt-level and instruction-level accuracy [16]. Their results show that attention-based steering improves instruction adherence while maintaining task performance, though excessive intervention can degrade output quality. This highlights the importance of calibrated control, which is particularly relevant for user-adaptive explanation systems where outputs must be tailored to different expertise levels.

Chapter 3

Methodology

3.1 Previous Work and Earlier Prototype

Before the current two-pipeline framework was developed, our earlier work focused on a modular ontology-grounded system for user-adaptive biomedical explanation generation across multiple tasks. The earlier framework was designed as a task-agnostic explanation architecture: for each biomedical task, a task-specific predictor was first used to obtain an output, after which local feature attribution, ontology-based grounding, and explanation generation were applied in sequence. The same overall explanation logic was reused across tasks, while the task model, dataset, and ontology resources varied depending on the biomedical setting.

Table 3.1: Biomedical tasks, datasets, models, and ontology resources used in the earlier prototype.

Dataset	Task Type	Finetuned Base Model	Ontology / KG Used
Medical Abstracts Text Classification Dataset	Multi-class Medical Abstract Classification	BioBERT	MeSH, DOID
NCBI Disease Corpus	Disease Named Entity Recognition	BioBERT	MeSH, UMLS
BioASQ (subset)	Biomedical Question Answering / Sentence Classification	BioBERT	MeSH
SymbiPredict	Disease Classification	BioBERT	MeSH, DOID, SYMP

The earlier prototype followed the workflow shown in Figure 3.1. Its stages can be summarized as follows:

1. **Task Model Prediction:** A fine-tuned biomedical classifier or task-specific predictor produces an output for the given input. Depending on the task, this may correspond to a disease class, named entity label, or answer span.
2. **Feature Importance Extraction:** Local feature attribution is computed using LIME or

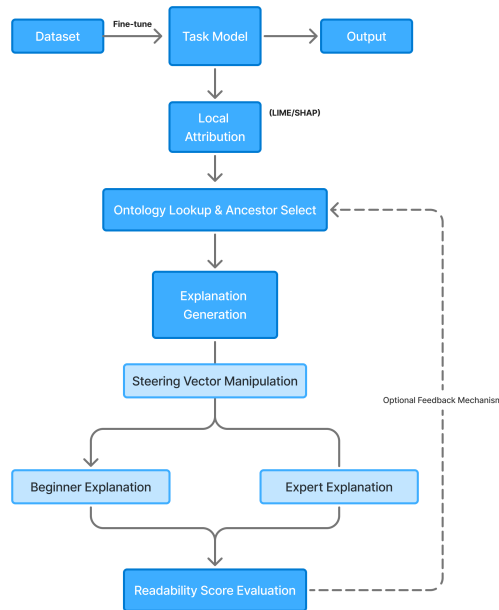


Figure 3.1: Earlier pipeline for ontology-grounded explanation generation.

SHAP in order to identify the most salient input tokens contributing to the model prediction.

3. **Ontology-Based Semantic Enrichment:** Salient tokens are grounded in biomedical ontologies such as MeSH, DOID, or UMLS. Relevant ancestors, semantic groupings, and concept abstractions are then retrieved to provide semantically meaningful explanatory context.
4. **Difficulty-Aware Ancestor Selection:** The ontology-derived concepts are filtered or abstracted according to the intended user type. Higher-level and simpler concepts are selected for beginner users, while deeper and more technical ancestors are retained for expert users.
5. **User-Adaptive Explanation Generation:** A transformer-based language model such as Gemma or BioBART receives the task output, salient features, ontology-grounded concepts, and user category, and generates a tailored natural-language explanation.
6. **Readability Score Evaluation:** The generated beginner and expert explanations are evaluated using readability metrics in order to verify whether the system successfully produces outputs of different linguistic complexity for different user groups.

This earlier system established the core idea that raw feature-based XAI signals can be enriched with biomedical ontology structure and verbalized into user-adaptive explanations, while also demonstrating applicability across multiple biomedical NLP tasks through task-specific pipeline instances. However, it had two key limitations: the pipelines remained task-specific and were not unified into a single framework, and user adaptiveness relied primarily on ontology abstraction and prompt-based generation without explicit inference-time control. In particular, mechanisms such as constrained

decoding were not incorporated, and steering vectors were not yet implemented but only identified as a potential direction. These limitations motivated the more focused and controlled design adopted in the current work.

3.2 Current System Overview

The present work refines the earlier prototype into two clearly defined pipelines for user-adaptive biomedical explanation generation. The first pipeline is an ontology-grounded explanation framework built around multi-class biomedical abstract classification, local feature attribution, semantic enrichment, and readability-constrained decoding. The second pipeline explores steering vectors as a complementary activation-level mechanism for controlling explanation difficulty.

The present implementation is centered on multi-class biomedical abstract classification as the primary ontology-grounded task, while steering-vector experiments are studied as a complementary difficulty-control pipeline. A fine-tuned BioBERT-based sequence classifier assigns each biomedical abstract to one of five disease category groups: Neoplasms, Digestive system diseases, Nervous system diseases, Cardiovascular diseases, and General pathological conditions. This focused scope provides a well-defined and reproducible evaluation surface, enabling rigorous comparison of readability and faithfulness metrics across audience tiers and ablation conditions.

3.3 Ontology-Grounded XAI + Constrained Decoding

3.3.1 Task Models and Datasets

This stage performs the underlying biomedical NLP task. Although the overall explainability framework is task-agnostic in principle, the present pipeline is instantiated specifically for multi-class biomedical abstract classification.

We use the **Medical Abstracts Text Classification Dataset**, a curated collection of 11,550 medical abstracts spanning five major disease-related categories. The class distribution is as follows:

- Neoplasms: 3163 abstracts
- Digestive system diseases: 1494 abstracts
- Nervous system diseases: 1925 abstracts
- Cardiovascular diseases: 3051 abstracts
- General pathological conditions: 4805 abstracts

The goal is to assign each abstract to exactly one disease category. This task is well-suited for evaluating the explanation pipeline because biomedical abstracts contain dense domain-specific terminology and naturally exhibit hierarchical medical concepts.

For ontology grounding, we use the **Disease Ontology (DOID)** and **MeSH (Medical Subject Headings)**. The rationale for this choice is:

- Both ontologies contain well-defined disease hierarchies aligned with the five categories in the dataset.
- DOID provides explicit parent–child disease relations useful for ancestor selection.
- MeSH includes synonyms, preferred labels, and hierarchical tree codes, facilitating robust ontology matching.
- Using disease-focused ontologies ensures semantically consistent grounding while avoiding unnecessary noise from broader knowledge sources.

These ontologies collectively provide high-quality semantic structures that support both shallow (beginner-friendly) and deep (expert-oriented) biomedical concept explanations.

3.3.2 Feature Importance Extraction

Once the classifier produces a prediction, the next stage identifies which input tokens were most influential in driving that decision. The pipeline implements a pluggable XAI attribution module supporting two complementary backends: Local Interpretable Model-Agnostic Explanations (LIME) [12] and Integrated Gradients (IG) [15]. Both methods produce a ranked list of (word, score) pairs that serve as anchors for downstream ontology grounding. The active method is selected via the `XAI_METHOD` configuration flag, enabling controlled ablation comparisons between the two attribution strategies.

LIME: Local Interpretable Model-Agnostic Explanations

LIME [12] is a widely adopted post-hoc interpretability method that explains individual predictions by approximating the classifier locally with an interpretable surrogate model. Given an input text x and a classifier f , LIME generates n perturbed variants of x by randomly masking subsets of tokens, queries f on each variant to obtain output probabilities, and fits a sparse linear model g to the resulting neighbourhood. The regression coefficients of g serve as attribution scores for individual tokens:

$$\text{LIME}(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

where $\mathcal{L}$ measures the fidelity of g to f in the local neighbourhood π_x (weighted by proximity to x), and $\Omega(g)$ is a complexity penalty that encourages sparsity [12]. In our implementation, a `LimeTextExplainer` is instantiated with $n = 300$ perturbation samples and the top- $k = 6$ features are returned for the predicted class. Multi-word biomedical entities identified by a NER pipeline are

fused with underscores prior to LIME so that clinically meaningful phrases (e.g., *cardiac tamponade*) are treated as single indivisible tokens rather than split across independent attribution slots.

The principal advantage of LIME in a biomedical context is that the returned feature tokens are surface-form words drawn directly from the input text, making them straightforwardly mappable to ontology concept labels without additional linguistic preprocessing. Its model-agnostic character also means the same attribution logic applies unchanged regardless of the underlying classifier architecture, ensuring that the explanation pipeline remains decoupled from the task model. LIME has been empirically validated to demonstrate high recall of interpretable model features, improve user trust judgement, aid model selection, and reveal spurious correlations in biomedical text classifiers [12, 1].

Integrated Gradients: Gradient-Based Token Attribution

Integrated Gradients (IG) [15] is a gradient-based attribution method grounded in axiomatic principles. It satisfies two fundamental desiderata: *sensitivity* (a feature that changes the prediction receives a non-zero attribution) and *implementation invariance* (attributions depend only on the input–output mapping, not the internal implementation). IG defines the attribution of the i -th input feature as the path integral of the model gradient with respect to that feature along a straight-line path from a neutral baseline $\bar{x}$ to the actual input x :

$$\text{IG}_i(x) = (x_i - \bar{x}_i) \times \int_{\alpha=0}^1 \frac{\partial F(\bar{x} + \alpha(x - \bar{x}))}{\partial x_i} d\alpha$$

In practice, the integral is approximated by summing gradients at $m = 50$ uniformly spaced interpolation steps between an all-zero embedding baseline $\bar{x}$ and the actual token embeddings x using the Captum library [5]. Attribution scores are computed at the embedding dimension level per token; per-token scalar scores are obtained by taking the L_2 norm of the attribution vector over the embedding dimension. Sub-word tokens produced by the BPE tokenizer (Qwen2.5 uses the `tiktoken` scheme) are aggregated back to whole-word level by summing their respective L_2 norms, and single-character artefacts are discarded. The top- k words by accumulated attribution magnitude are returned.

Relative to LIME, IG provides attributions that are model-intrinsic — tied to the actual gradient of the classifier with respect to its input representation — rather than surrogate-approximated, and are therefore more faithful to the model’s internal computation by construction [15]. However, because IG operates at the embedding level, the returned tokens are subword-aggregated reconstructions rather than verbatim surface spans, which can occasionally introduce minor string normalisation artefacts at the ontology lookup stage. The two methods are offered as interchangeable backends, ensuring that downstream ontology enrichment and constrained explanation generation remain agnostic to the chosen attribution strategy.

3.3.3 Ontology-Based Semantic Enrichment

The salient terms identified by the attribution module are enriched with biomedical semantics. This enrichment step consists of:

- case normalization and lemmatization,
- synonym expansion (ontology labels),
- ID-based and string-based concept lookup in MeSH, DOID, UMLS, and CTD,
- retrieving:
 - preferred names,
 - hierarchical ancestors,
 - semantic relations,
 - disease–symptom or disease–chemical links,
 - high-level ontological groupings.

This forms a structured interpretation layer that captures biomedical meaning beyond raw tokens.

Ontology Selection Strategy

Biomedical ontologies can be selected dynamically at inference time using federated services such as the **BioPortal API**, which provides programmatic access to ontology classes, synonyms, and mappings across hundreds of biomedical knowledge sources. However, to ensure reproducibility and computational stability, our current prototype adopts a simplified strategy: for each task, we manually select the most relevant ontology (e.g., MeSH or DOID for disease classification, CTD for chemical–disease relations, UMLS for NER). These ontologies are downloaded locally and loaded through `owlready2` for fast lookup and ancestor extraction.

This controlled setup allows consistent scoring, depth computation, and stability in the user-adaptive difficulty model, while retaining compatibility with future integration of real-time BioPortal querying in a unified multi-task setting.

Extracting Hierarchical Ancestors

For each mapped concept, we extract a chain of ancestors (general $\rightarrow$ specific). However, not all ancestors are equally interpretable, motivating the need for difficulty modelling.

3.3.4 Difficulty Modeling

This stage forms the **first level of user adaptiveness**. We adopt the difficulty model of Venugopal and Kumar [17], which treats interpretability as a function of:

- **Specificity**: how deep or technical a concept is,
- **Popularity**: how frequently the concept appears in ontology relations.

Specificity

Defined as:

$$S(c) = \frac{\text{depth}(c)}{\text{max_depth}}$$

Higher specificity implies deeper and more technical concepts. This aligns with findings from [17] that deeper nodes impose greater cognitive load.

Popularity

Defined as:

$$P(c) = \frac{\sum_{r \in R} \mathbf{1}(c \in \text{range}(r) \cup \text{domain}(r))}{|C|}$$

More popular concepts tend to be easier for non-experts.

Score Formulation

We assign difficulty-aware scores as:

$$\text{score}_{\text{beginner}}(c) = 0.6P(c) + 0.4(1 - S(c))$$

$$\text{score}_{\text{expert}}(c) = S(c)$$

$$\text{score}_{\text{intermediate}}(c) = 1 - |S(c) - 0.5|$$

These reflect cognitive preferences for different users:

- Beginners prefer broad, familiar concepts.
- Experts prefer deep, technical concepts.
- Intermediate users prefer mid-level concepts.

This scoring determines which ontology ancestors are passed to the explanation model.

3.3.5 User-Adaptive Explanation Generation

The final stage synthesizes all upstream information — the classifier prediction, XAI feature attributions, ontology-derived ancestor chains, and an explicit audience category flag — into a coherent natural-language explanation using Qwen2.5-1.5B-Instruct [11]. This stage implements two complementary mechanisms for user adaptation: (i) a prompt-level soft instruction that encodes the audience description, and (ii) a generation-level hard constraint enforced through readability-aware beam decoding.

Readability-Constrained Beam Decoding

Prompt-based audience conditioning alone is insufficient to enforce lexical simplicity: empirical evaluation shows that without generation-time constraints, the model produces vocabulary of comparable complexity for beginner and expert audiences. To address this limitation, we implement *readability-constrained beam search* by integrating a custom `ReadabilityLogitsProcessor` into the Hugging Face `generate()` API [19]. This processor intercepts the raw logit distribution at every decoding step and applies a token-level penalty that suppresses vocabulary associated with linguistic difficulty, steering the model toward simpler language without any gradient update or model retraining. This makes the approach model-agnostic and computationally lightweight: a single Qwen2.5 checkpoint serves all three audience tiers, differentiated solely by the penalty strength parameter λ .

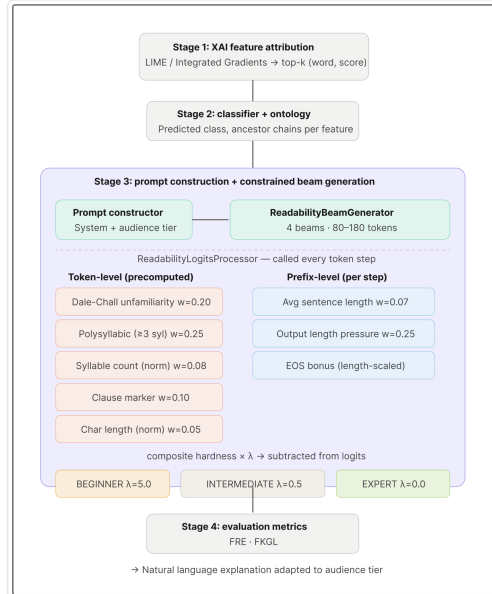


Figure 3.2: Overview of the readability-constrained beam decoding pipeline.

Vocabulary-Level Feature Precomputation At initialisation time, the processor iterates over every token in the Qwen2.5 tokenizer vocabulary (32,000 tokens) and precomputes five fixed feature tensors:

- **Dale-Chall unfamiliarity** (`token_dale_chall`): a binary flag set to 1.0 for tokens absent from the Dale-Chall familiar-word list, augmented with biomedical layperson vocabulary.
- **Syllable count** (`token_syllable`): estimated syllable count normalised by a soft cap of 4.0, clamped to $[0, 1]$.
- **Polysyllabic flag** (`token_polysyllabic`): a binary indicator set to 1.0 for tokens with ≥ 3 syllables.
- **Clause marker** (`token_clause`): a binary flag for clause-introducing connectives.
- **Character length** (`token_char_len`): normalised character count per token, clamped to $[0, 1]$ with a soft cap of 9 characters.

Only pure alphabetic token pieces are assigned non-zero feature values; punctuation, numbers, and special tokens are explicitly excluded. This precomputation is performed once, making per-step penalty application a single vectorised subtraction over the vocabulary.

Composite Hardness Signal At each decoding step, the five token-level features are combined with two prefix-level features — text length pressure and average sentence length — into a composite hardness score $h(t)$:

$$h(t) = \sum_{i=1}^7 w_i \cdot h_i(t), \quad \text{with } \sum_{i=1}^7 w_i = 1.0$$

The configured weights and their readability formula motivations are summarised in Table 3.2.

Table 3.2: Hardness feature weights and corresponding readability formula sources.

Feature	Weight	Formula Source
Polysyllabic flag (≥ 3 syllables)	0.25	SMOG Index; Gunning Fog
Text length pressure	0.25	Design heuristic (conciseness)
Dale-Chall unfamiliarity	0.20	Dale-Chall Formula
Clause marker density	0.10	Syntactic complexity
Syllable count (normalised)	0.08	Flesch Reading Ease
Average sentence length	0.07	Flesch-Kincaid Grade Level
Average characters per word	0.05	Automated Readability Index

Penalty Application and Audience Scaling At each decoding step, the adjusted logit for token t is computed as:

$$\hat{s}(t) = s(t) - \lambda \cdot h(t)$$

where $s(t)$ is the raw log-probability score produced by the model and λ is an audience-specific penalty strength. For the EOS token specifically, the penalty is inverted to a bonus proportional to the prefix-level length pressure:

$$\hat{s}(\text{EOS}) = s(\text{EOS}) + \lambda \cdot (w_{\text{len}} \cdot h_{\text{len}} + w_{\text{sent}} \cdot h_{\text{sent}})$$

The λ values are configured per audience tier as shown in Table 3.3.

Table 3.3: Audience-specific penalty strength (λ) values governing constrained generation.

Audience	λ	Effect
BEGINNER	5.0	Strong suppression of complex vocabulary and long sentences
INTERMEDIATE	0.5	Moderate simplification; domain vocabulary permitted
EXPERT	0.0	No penalty; standard beam search

Beam Search Configuration and Design Rationale The constrained decoder is paired with beam search (4 beams) rather than temperature-based sampling. Beam search evaluates entire candidate trajectories simultaneously, so beams that accumulate multiple penalised complex tokens suffer compounding score reductions and are progressively pruned in favour of beams that consistently select simpler vocabulary.

Dual-Channel Adaptation and Ablation The constrained decoding mechanism operates in addition to, not in place of, the prompt-level audience instruction. The system prompt instructs the model to adapt its register to the specified audience tier; the logit processor then acts as a hard enforcement layer that suppresses complex tokens regardless of the model’s base generation tendency. This dual-channel design — soft prompt guidance plus hard penalty constraint — is empirically more robust than either mechanism in isolation.

3.4 Steering Vectors

3.4.1 Contrastive Dataset Construction

The steering pipeline begins with the construction of a contrastive dataset of easier and harder explanations. As a source of diverse explanatory language, we use a subset of the **Reddit Explain Like I’m 5 (ELI5)** dataset. An initial set of 500 samples is collected and stored as `eli5_500.csv`. From this broader pool, a filtered subset is curated for steering-vector learning.

For each selected question, three stylistic variants are generated using an Ollama-hosted instruction model **gemma/2b**:

- **Beginner:** very simple words and short sentences,
- **Intermediate:** clear but moderately detailed language,
- **Expert:** detailed and technical explanation style.

The readability of each generated explanation is measured using Flesch Reading Ease (FRE) and Flesch–Kincaid Grade Level (FKGL). Only examples that exhibit clear readability separation across the three levels are retained. The filtering conditions are:

- **Beginner:** $\text{FRE} \geq 70$ and $\text{FKGL} \leq 6$,
- **Intermediate:** $40 \leq \text{FRE} \leq 70$ and $6 \leq \text{FKGL} \leq 10$,
- **Expert:** $\text{FRE} \leq 50$ and $\text{FKGL} \geq 10$.

This produces a filtered dataset of approximately 50 high-quality contrastive examples. To verify that the retained samples exhibit strong stylistic separation, Cohen’s d is computed between beginner and expert outputs for both FRE and FKGL. The resulting large effect sizes confirm that the contrastive pairs encode a meaningful difficulty distinction.

Although three levels are generated during this curation stage, the steering vector itself is learned only from the *easy* versus *hard* extremes, where:

- **easy** = beginner output,
- **hard** = expert output.

These pairs are stored in `steering_pairs.json` and `steering_pairs.csv`.

3.4.2 Steering Vector Extraction

Steering-vector extraction is performed using **microsoft/phi-2**, which serves as the base model for representation learning and subsequent steered generation. Let $h_\ell(x)$ denote the hidden-state representation at transformer layer ℓ for input text x . In our implementation, we extract hidden states from an intermediate layer (layer 16), which provides a balance between low-level lexical representation and high-level semantic structure.

For each contrastive pair $(x_{\text{easy}}^{(i)}, x_{\text{hard}}^{(i)})$, the hidden-state tensor is mean-pooled across sequence positions to obtain a single fixed-dimensional vector per explanation:

$$r_{\text{easy}}^{(i)} = \text{Pool}\left(h_\ell(x_{\text{easy}}^{(i)})\right), \quad r_{\text{hard}}^{(i)} = \text{Pool}\left(h_\ell(x_{\text{hard}}^{(i)})\right)$$

where $\text{Pool}(\cdot)$ denotes mean pooling over tokens. The average easy and hard representations are then computed as:

$$\mu_{\text{easy}} = \frac{1}{N} \sum_{i=1}^N r_{\text{easy}}^{(i)}, \quad \mu_{\text{hard}} = \frac{1}{N} \sum_{i=1}^N r_{\text{hard}}^{(i)}$$

The steering direction is defined as the normalized centroid difference:

$$v_{\text{steer}} = \frac{\mu_{\text{easy}} - \mu_{\text{hard}}}{\|\mu_{\text{easy}} - \mu_{\text{hard}}\|}$$

This vector captures a latent direction in activation space corresponding to a shift from harder explanations toward easier ones. In the implementation, the learned direction together with the two centroids are saved as `v_steering.npy`, `mu_easy.npy`, and `mu_hard.npy`.

3.4.3 Activation-Space Intervention

Once extracted, the steering vector is injected into the same base model during generation. Let $h_{\ell}^{(t)}$ denote the hidden state at decoding step t for layer ℓ . The steered hidden state is computed as:

$$\hat{h}_{\ell}^{(t)} = h_{\ell}^{(t)} - \alpha v_{\text{scaled}}$$

where α is a tunable steering coefficient and v_{scaled} is the steering vector after norm-based rescaling. Because the underlying direction is defined as $\mu_{\text{easy}} - \mu_{\text{hard}}$, negative values of α move the generation toward easier language, while positive values push it toward harder and more technical explanations.

In practice, steering is implemented using a forward hook registered on layer 16 of `phi-2`. To stabilise the intervention, the hidden-state norm of a reference input is first measured, and the steering vector is scaled by a small fraction of this norm. In our implementation, a scale fraction of 0.05 is selected as a stable operating point after initial experimentation. This avoids overly aggressive perturbations that could degrade fluency or coherence.

Generation itself is performed with sampling-based decoding using `phi-2`, with a temperature of 0.7, top- p of 0.9, repetition penalty of 1.1, and a maximum length of 120 new tokens. Only the generated continuation is decoded, excluding the original prompt prefix.

3.4.4 Alpha Sweep

To calibrate the steering strength, we perform an alpha sweep over a range of values from negative to positive directions. For each value of α , the steered model generates explanations for a small set of representative prompts, and the resulting outputs are evaluated using FRE and FKGL. The trend is shown in Figure 4.6.

The alpha sweep confirms that the learned steering direction corresponds meaningfully to explanation difficulty: more negative values produce easier explanations, while more positive values produce more difficult and technical outputs. Based on this calibration, we select the following steering levels for downstream biomedical evaluation:

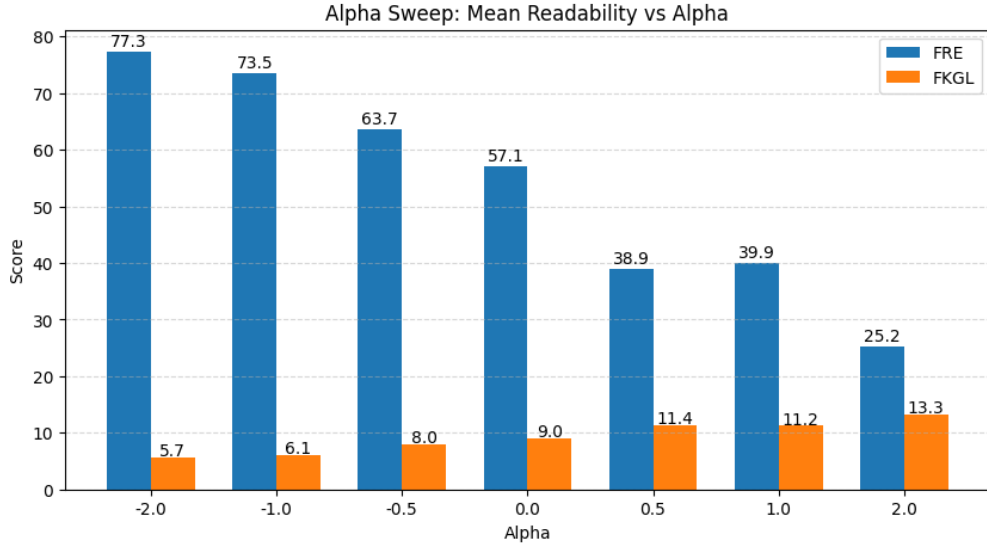


Figure 3.3: Mean readability scores across steering coefficients. Negative α values yield simpler explanations with higher FRE and lower FKGL, while positive values produce more complex outputs.

$$\text{LEVELS} = \begin{cases} \text{beginner} = -1.0 \\ \text{intermediate} = 0.0 \\ \text{expert} = 2.0 \end{cases}$$

Although $\alpha = -2.0$ produced the highest readability, we selected $\alpha = -1.0$ for the beginner setting in order to avoid over-simplification and preserve better semantic fidelity. The neutral point $\alpha = 0.0$ is used for intermediate outputs, while $\alpha = 2.0$ is retained for expert explanations because it consistently produces the strongest increase in technical complexity.

3.4.5 Biomedical Evaluation Setup

After the steering vector is learned from ELI5-derived contrastive examples, it is evaluated on biomedical text generation. A small fixed set of six biomedical passages is used as the evaluation set, including examples from disease classification and biomedical abstract text. For each biomedical input, the model generates three explanations corresponding to beginner, intermediate, and expert steering levels. These outputs are then evaluated using readability metrics, primarily FRE and FKGL.

To isolate the effect of steering, the steered outputs are also compared against a prompt-only baseline. In the baseline condition, the same model generates beginner and expert explanations from explicit textual prompts alone, without activation intervention. The comparison between prompt-only and steered outputs is summarised using readability tables and plots, allowing us to measure whether latent activation control produces clearer user separation than prompt conditioning alone.

3.5 Evaluation Metrics

Explanation quality is evaluated along two axes: **readability** and **faithfulness**.

3.5.1 Readability

To measure linguistic complexity across different user modes, we compute:

- Flesch Reading Ease (FRE),
- Flesch–Kincaid Grade Level (FKGL).

These metrics capture lexical difficulty, syntactic complexity, and sentence structure.

3.5.2 Faithfulness

Faithfulness captures how well the explanation aligns with true model behaviour and ontology facts. We use:

1. **LIME Coverage:** The proportion of high-weight LIME tokens referenced directly or indirectly in the explanation.
2. **Ontology Support:** Verifies that biomedical claims in the explanation are grounded in MeSH, DOID, CTD, or UMLS.

We justify the use of LIME based on established validation benchmarks from prior work: LIME demonstrates high recall of interpretable model features, improves user trust judgment, aids model selection, and reveals spurious correlations.

These metrics provide a principled starting point for comparing ontology-grounded constrained decoding and steering-vector-based control of explanation style. float

Chapter 4

Results

This chapter presents both the findings from our earlier ontology-grounded prototype and the new experiments introduced in the current work. We first summarise the key results obtained from the previous multi-task framework, including biomedical classification and question answering. We then present new results for two inference-time difficulty-control mechanisms: readability-constrained decoding and steering vectors.

4.1 Results from Previous Work

The earlier ontology-grounded prototype described in Chapter 3 was evaluated across multiple biomedical NLP tasks. Its purpose was to test whether ontology-based semantic enrichment and user-aware explanation prompting could produce different explanation styles for different audiences. In this section, we briefly summarise those earlier findings, which motivated the more focused constrained decoding and steering-vector experiments introduced later in this chapter.

4.1.1 Multi-Class Medical Abstract Classification

This subsection summarises the explanation-level evaluation conducted in the earlier prototype on the multi-class biomedical abstract classification pipeline. The objective was to analyse how explanation complexity varied across user categories and ontology configurations.

Experimental Setup

In the earlier prototype, 50 random samples from the Medical Abstracts Text Classification Dataset were passed through the ontology-grounded explanation pipeline under four conditions:

- **Beginner (Normal Mode)** — user-adaptive ancestor selection enabled
- **Expert (Normal Mode)** — user-adaptive ancestor selection enabled
- **Beginner (Full Mode)** — all ontology ancestors included

- **Expert (Full Mode)** — all ontology ancestors included

This design allowed us to disentangle:

1. the effect of user category,
2. the effect of ontology depth,
3. the interaction between the two.

Readability metrics were computed for every generated explanation and averaged across samples.

Readability Comparison Across User Types

Readability was quantified using:

- **Flesch Reading Ease (FRE)** — higher values indicate simpler text
- **Flesch–Kincaid Grade Level (FKGL)** — higher values indicate more complex text

Normal Mode (Adaptive Ancestor Selection). Table 4.1 summarises readability differences when only the top- k ontology ancestors were used.

User Category	FRE ($\uparrow$)	FKGL ($\downarrow$)
Beginner	31.23	12.04
Expert	8.60	16.78

Table 4.1: Readability metrics for Normal Mode (top- k ancestor selection) in the earlier prototype.

A clear separation appears between user categories:

- Beginner explanations are substantially simpler.
- Expert explanations are more technical and require a higher reading level.

This confirms that the difficulty-aware ancestor selection mechanism effectively modulates explanation complexity.

Conclusion (Normal Mode). The readability gap demonstrates that the earlier ontology-grounded pipeline successfully tailored explanations according to user category when ontology depth selection was enabled.

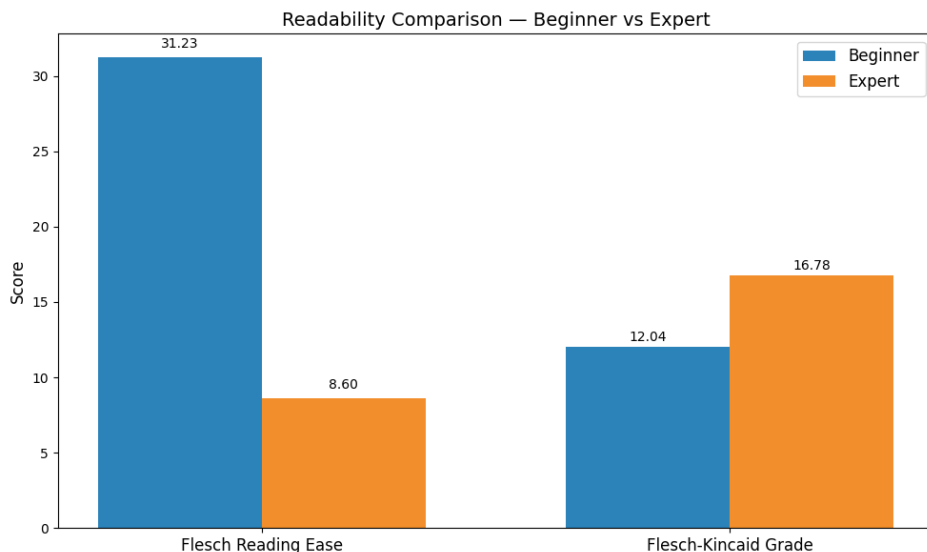


Figure 4.1: Beginner vs. Expert readability comparison in Normal Mode.

User Category	FRE ($\uparrow$)	FKGL ($\downarrow$)
Beginner	14.31	16.15
Expert	12.84	16.00

Table 4.2: Readability metrics for Full Mode (all ontology ancestors) in the earlier prototype.

Ablation: Full Ontology Mode. To isolate the role of ontology depth, ancestor selection was disabled and the LLM was provided with *all* ancestors associated with each biomedical concept. Table 4.2 shows the readability metrics.

Unlike Normal Mode, the two user categories produce very similar readability scores. This occurs because:

- deeper and more numerous ancestors dominate the prompt,
- the LLM is forced to use technical biomedical language,
- user-adaptive cues become overshadowed by dense ontology content.

Conclusion (Full Mode). Providing the full ancestor list removes effective user adaptiveness. Both beginner and expert explanations become similarly technical, highlighting that **ancestor selection is essential** for controlling explanation difficulty.

Overall Finding. Taken together, these results showed that ontology-based ancestor selection was a key driver of user adaptiveness in the earlier prototype, but they did not yet incorporate any explicit decoding-time control over explanation complexity.

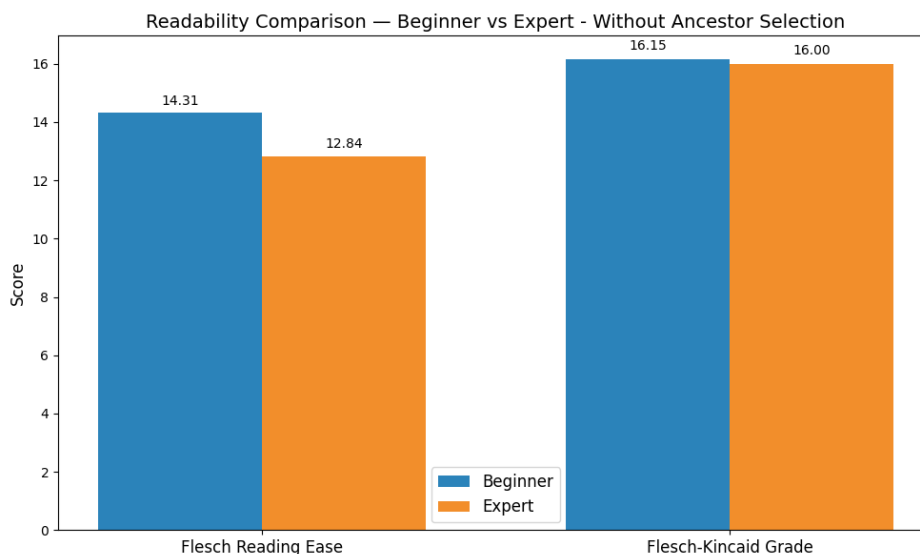


Figure 4.2: Readability comparison in Full Mode (all ontology ancestors).

4.1.2 Biomedical Question Answering (BioASQ Subset)

This subsection summarises explanation behaviour in the earlier prototype for a biomedical question answering (QA) system applied to a subset of the BioASQ 6b benchmark. As in the previous task, the focus was not model accuracy but the behaviour of the explanation-generation pipeline across user categories and ontology configurations.

Experimental Setup

The QA system was evaluated on the `test-factoid-6b-1.json` set, which contains 121 factoid-style biomedical questions. A finetuned **BioBERT QA model** (`dmis-lab/biobert-base-cased-v1.1-squad`) was used to extract answer spans from the provided context passages. To support structured explanation generation, predicted answers and influential context terms were grounded in the **MeSH** biomedical ontology.

For each QA sample, explanations were generated under four settings:

- **Beginner (Plain)** — LLM receives only question, answer, and context
- **Expert (Plain)** — same as above, but phrased for experts
- **Beginner (Ontology)** — MeSH concepts and ancestor definitions included
- **Expert (Ontology)** — ontology-aware expert explanation

Two representative BioASQ samples were used for controlled comparison, and readability metrics were averaged across them.

Readability Comparison Across User Types

Readability was quantified using:

- **Flesch Reading Ease (FRE)** — higher values indicate simpler text
- **Flesch–Kincaid Grade Level (FKGL)** — higher values indicate more complex text

Explanations Without Ontology. Table 4.3 summarises readability when explanations were generated without ontology information.

User Category	FRE (↑)	FKGL (↓)
Beginner	72.60	6.03
Expert	-1.85	18.66

Table 4.3: Readability metrics for QA explanations without ontology grounding.

A strong separation emerges:

- Beginner explanations are highly readable and accessible.
- Expert explanations become extremely technical (negative FRE indicates dense biomedical phrasing).

This shows that even without ontology, the prompting mechanism induces strong user adaptiveness.

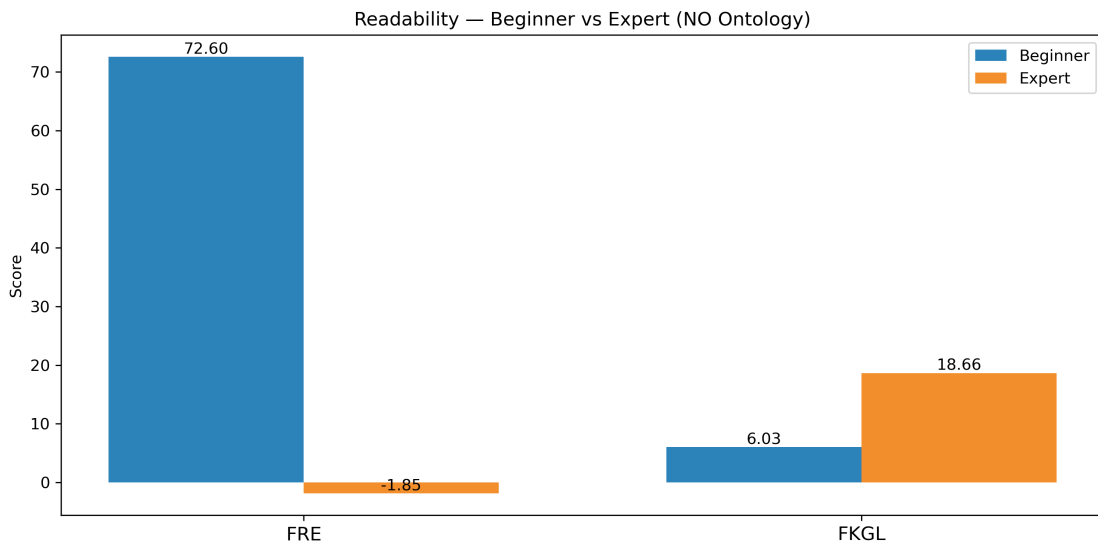


Figure 4.3: Beginner vs. Expert readability comparison for QA (no ontology).

Conclusion (No Ontology). Expert explanations are significantly more complex than beginner explanations, but the difference is unregulated, leading to extremely technical expert outputs.

Ontology-Grounded Explanations. Table 4.4 shows readability when MeSH concepts and ancestors were included.

User Category	FRE ($\uparrow$)	FKGL ($\downarrow$)
Beginner	39.85	12.85
Expert	12.06	17.26

Table 4.4: Readability metrics for QA explanations with ontology grounding.

Key observations include:

- Both beginner and expert explanations become more complex than in the plain setting.
- The readability gap narrows once ontology content is introduced.
- Ontology definitions introduce unavoidable domain-specific terminology that raises baseline difficulty.

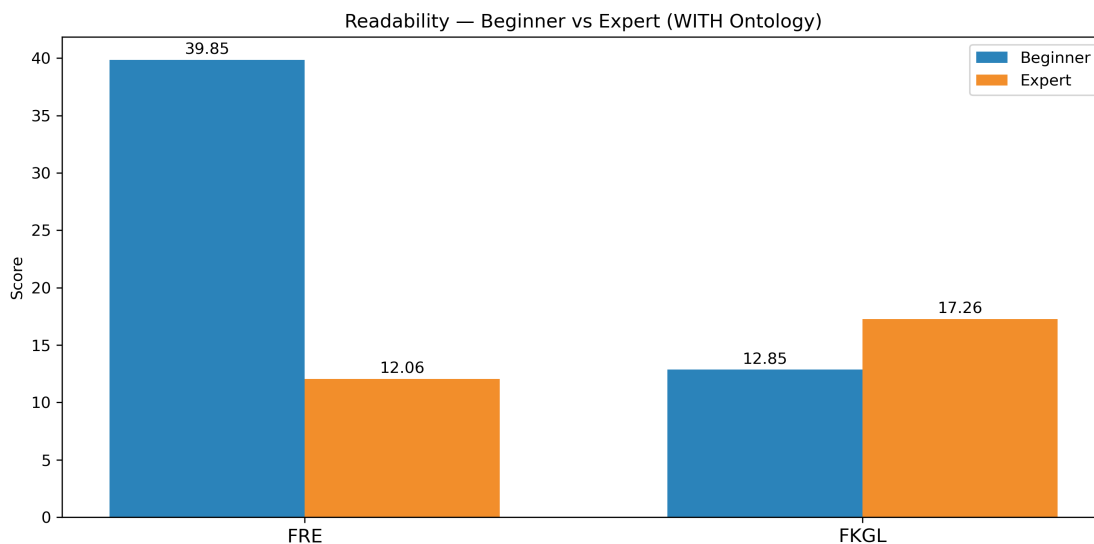


Figure 4.4: Readability comparison for QA explanations with ontology grounding.

Conclusion (Ontology). Ontology grounding introduces consistent biomedical terminology across user types, raising the overall complexity of explanations while still preserving distinctions between beginner and expert phrasing.

Overall Finding. Taken together, the QA experiments confirmed that user-adaptive explanation behaviour could be induced through prompting and ontology grounding, but also showed that ontology content could substantially raise explanation complexity, especially for beginner users.

4.1.3 Summary of Previous Findings

The earlier ontology-grounded prototype established three key findings. First, explanation difficulty can be modulated through ontology-aware concept selection, particularly in the multi-class classification setting. Second, ontology grounding improves semantic consistency but often increases baseline textual complexity, especially in question answering. Third, prompt-level adaptation alone is insufficient to provide fine-grained control over readability. These observations directly motivated the introduction of explicit inference-time control mechanisms in the current work, namely constrained decoding and steering vectors.

Mode / User Type	Setting	FRE ($\uparrow$)	FKGL ($\downarrow$)
Beginner	Normal (Adaptive)	31.23	12.04
Expert	Normal (Adaptive)	8.60	16.78
Beginner	Full Ontology	14.31	16.15
Expert	Full Ontology	12.84	16.00
Beginner	QA Plain (No Ontology)	72.60	6.03
Expert	QA Plain (No Ontology)	-1.85	18.66
Beginner	QA + Ontology	39.85	12.85
Expert	QA + Ontology	12.06	17.26

Table 4.5: Unified readability comparison across all previous-work evaluation modes.

4.2 Effect of Constrained Decoding on Explanation Readability

Building on the earlier ontology-grounded prototype, we next evaluate readability-constrained decoding as an explicit generation-time mechanism for controlling explanation difficulty. Unlike the previous results, which relied primarily on ontology abstraction and prompt-level conditioning, this experiment tests whether decoding-time penalties can directly improve readability while preserving user separation.

4.2.1 Experimental Setup

The constrained decoding ablation was carried out on the multi-class medical abstract classification task. Five randomly selected samples from the Medical Abstracts Text Classification Dataset were processed under two conditions:

- **With CD** — explanation generation guided by the readability-aware penalty
- **Without CD** — standard generation with no readability penalty (baseline)

Each sample was evaluated for both **beginner** and **expert** audience tiers, yielding four cells in the comparison matrix. The objective was to isolate the contribution of constrained decoding beyond ontology grounding and prompt-level user adaptation.

4.2.2 Results

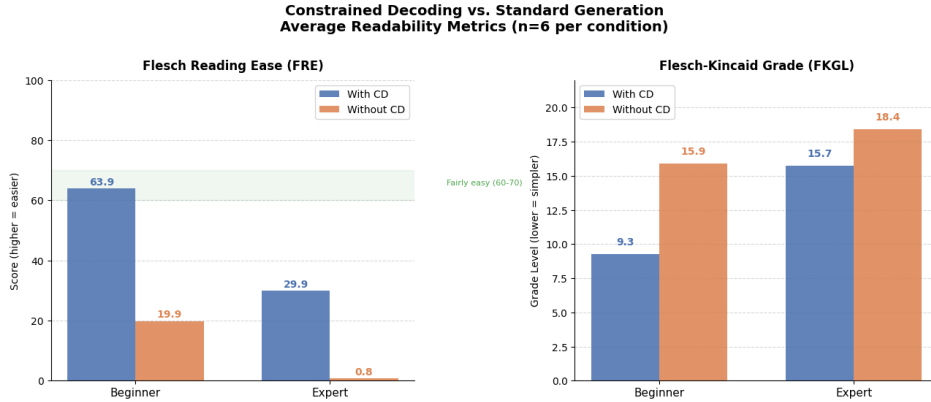


Figure 4.5: Average Flesch Reading Ease (FRE) and Flesch–Kincaid Grade Level (FKGL) with and without constrained decoding, broken down by audience tier. Higher FRE indicates simpler text; lower FKGL indicates a lower required reading grade.

Table 4.6 summarises the average readability metrics across both conditions.

Audience	FRE (With CD)	FRE (No CD)	Δ FRE	FKGL (With CD)	FKGL (No CD)	Δ FKGL
Beginner	63.90	19.88	+44.03	9.28	15.92	−6.64
Expert	29.92	0.76	+29.16	15.74	18.40	−2.65

Table 4.6: Readability comparison with and without constrained decoding ($n = 5$ samples per cell). Δ values represent the gain attributable to constrained decoding (positive Δ FRE and negative Δ FKGL indicate improvement).

4.2.3 Discussion

The constrained decoding results show that explicit readability control substantially improves explanation accessibility for both audience tiers while preserving the intended distinction between them. For beginner users, enabling constrained decoding raises FRE from 19.88 to 63.90 and reduces FKGL from 15.92 to 9.28, demonstrating a large shift from difficult, academically dense text to substantially more accessible explanations. For expert users, the gains are smaller but still meaningful, with FRE increasing from 0.76 to 29.92 and FKGL decreasing from 18.40 to 15.74, indicating that the penalty softens unnecessarily dense prose without eliminating technical depth. Overall, these findings confirm that constrained decoding is an effective and lightweight inference-time mechanism for calibrating explanation difficulty: it requires no additional fine-tuning, introduces only modest runtime overhead, and integrates naturally with the ontology-grounded explanation pipeline.

4.3 Effect of Steering Vectors on Explanation Readability

We next evaluate steering vectors as a latent activation-based mechanism for controlling explanation difficulty. Unlike constrained decoding, which acts explicitly on token probabilities, steering vectors

modify the internal hidden-state trajectory of the language model during generation. The goal of this experiment is to determine whether activation-space intervention can reliably separate beginner, intermediate, and expert explanation styles on biomedical text.

4.3.1 Experimental Setup

The steering-vector evaluation was performed on five biomedical samples drawn from the Medical Abstracts Text Classification Dataset. A steering vector was first learned from a filtered subset of Reddit ELI5 contrastive explanation pairs, where easy and hard explanations were generated using an Ollama-hosted Gemma model and then used to derive a latent steering direction in `microsoft/phi-2`. During evaluation, `phi-2` generated explanations for the biomedical samples under three steering levels:

- **Beginner:** $\alpha = -1.0$
- **Intermediate:** $\alpha = 0.0$
- **Expert:** $\alpha = 2.0$

To isolate the contribution of steering, the same biomedical inputs were also evaluated under a prompt-only baseline without activation intervention. Explanation quality was measured using Flesch Reading Ease (FRE) and Flesch–Kincaid Grade Level (FKGL).

4.3.2 Results

We first calibrated the steering coefficient by sweeping α across a range of negative and positive values. For each value, the steered model generated explanations for a small set of representative prompts, and the resulting outputs were evaluated using FRE and FKGL. The resulting trend is shown in Figure 4.6.

The alpha sweep confirms that the learned steering direction corresponds meaningfully to explanation difficulty: more negative values produce easier explanations, while more positive values produce more difficult and technical outputs. Based on this calibration, we selected the following steering levels for downstream biomedical evaluation:

$$\text{LEVELS} = \begin{cases} \text{beginner} = -1.0 \\ \text{intermediate} = 0.0 \\ \text{expert} = 2.0 \end{cases}$$

Although $\alpha = -2.0$ produced the highest readability, we selected $\alpha = -1.0$ for the beginner setting in order to avoid over-simplification and preserve better semantic fidelity. The neutral point $\alpha = 0.0$ is used for intermediate outputs, while $\alpha = 2.0$ is retained for expert explanations because it consistently produces the strongest increase in technical complexity.

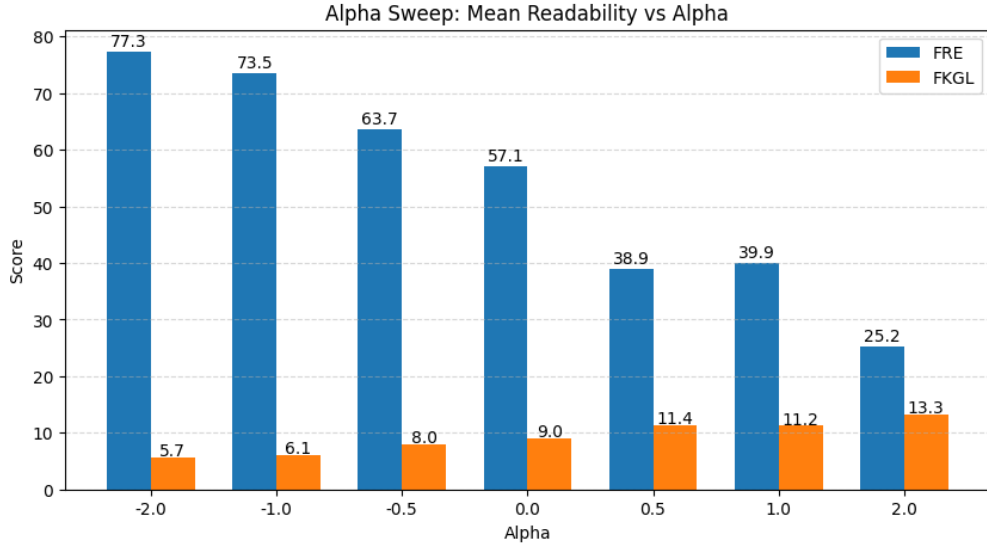


Figure 4.6: Mean readability scores across steering coefficients. Negative α values yield simpler explanations with higher FRE and lower FKGL, while positive values produce more complex outputs.

After fixing these steering levels, we evaluated the resulting beginner, intermediate, and expert explanations on the biomedical passages. Figure 4.7 summarises the average readability scores across the selected steering settings.

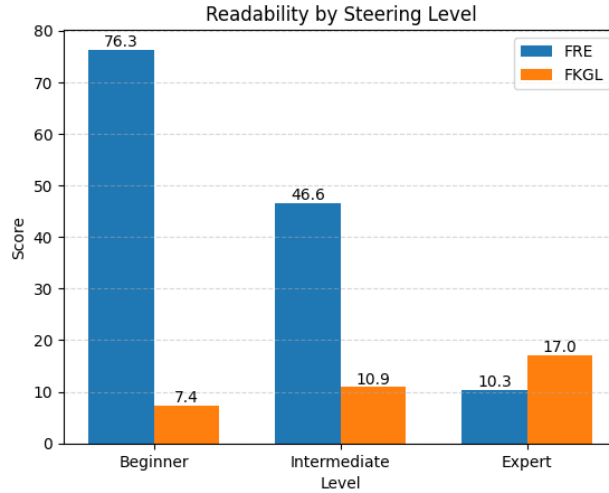


Figure 4.7: Mean readability scores across the selected steering levels.

Table 4.7 summarises the average readability scores across the five biomedical evaluation samples. The selected steering levels exhibit a clear monotonic separation in readability. Beginner steering produces the simplest outputs, with a high FRE of 76.3 and a low FKGL of 7.4. Intermediate steering occupies a stable midpoint, while expert steering yields much denser and more technical explanations, with FRE dropping to 10.3 and FKGL rising to 17.0. This confirms that the learned steering direction can be used as a practical control axis for explanation difficulty.

To determine whether steering provides stronger audience separation than prompting alone, we

Steering Level	FRE ($\uparrow$)	FKGL ($\downarrow$)
Beginner	76.3	7.4
Intermediate	46.6	10.9
Expert	10.3	17.0

Table 4.7: Average readability metrics for biomedical explanations generated at the selected steering levels.

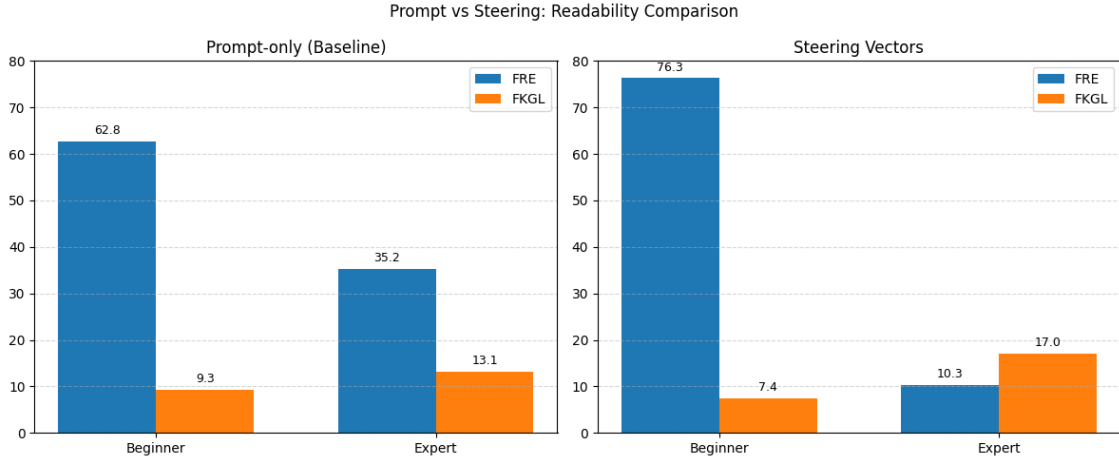


Figure 4.8: Prompt-only versus steering-vector readability comparison for beginner and expert biomedical explanations. Steering yields a stronger separation between audience levels than prompt conditioning alone.

compare the steered outputs against a prompt-only baseline in Figure 4.8.

Table 4.8 presents the corresponding mean readability values.

Method	Audience	FRE ($\uparrow$)	FKGL ($\downarrow$)
Prompt-only	Beginner	62.8	9.3
Prompt-only	Expert	35.2	13.1
Steering Vectors	Beginner	76.3	7.4
Steering Vectors	Expert	10.3	17.0

Table 4.8: Prompt-only baseline versus steering-vector readability comparison for biomedical explanations.

Compared with prompt-only conditioning, steering produces a stronger readability separation between beginner and expert outputs. In the beginner case, steering improves readability by increasing FRE from 62.8 to 76.3 and reducing FKGL from 9.3 to 7.4. In the expert case, steering pushes the explanation toward a substantially more technical register, lowering FRE from 35.2 to 10.3 and increasing FKGL from 13.1 to 17.0. This indicates that activation-space intervention offers a more powerful mechanism for explicit difficulty modulation than prompt wording alone.

4.3.3 Discussion

The steering-vector results show that latent activation control can reliably modulate biomedical explanation difficulty without retraining the underlying language model. The alpha sweep confirms that the learned direction corresponds meaningfully to readability, and the final mapping of $\alpha =$

−1.0, 0.0, and 2.0 produces a clear progression from beginner to intermediate to expert outputs. At these selected levels, beginner explanations achieve a high FRE of 76.3 and low FKGL of 7.4, whereas expert explanations drop to an FRE of 10.3 and rise to an FKGL of 17.0, with intermediate outputs occupying a stable midpoint. The comparison with the prompt-only baseline further shows that steering yields stronger audience separation than prompt wording alone, especially by making beginner explanations more readable and expert explanations more distinctly technical. At the same time, steering remains an implicit control mechanism operating in representation space rather than through transparent output-level constraints, which makes it flexible and continuous but less directly interpretable than constrained decoding. Thus, steering vectors are best viewed as a complementary mechanism for difficulty modulation rather than a replacement for ontology-grounded explanation generation.

4.4 Comparative Discussion

Taken together, the results reveal three stages in the evolution of the user-adaptive explanation framework. The earlier ontology-grounded prototype showed that explanation difficulty could be influenced through ontology selection and prompting, but this control remained indirect and task-specific. Constrained decoding extends that framework by adding explicit token-level readability control, producing strong improvements in accessibility while remaining tightly aligned with ontology-grounded explanation generation. Steering vectors, in contrast, provide a latent and continuous control mechanism that can shift explanation difficulty through hidden-state intervention and outperform prompt-only conditioning in readability separation. From a practical standpoint, constrained decoding offers stronger interpretability and clearer readability guarantees, whereas steering vectors offer greater flexibility and smoother style control. These findings suggest that explicit symbolic control and latent activation-based control should be treated as complementary strategies: the ontology-grounded pipeline provides semantic reliability, while constrained decoding and steering vectors provide two different mechanisms for adapting explanation difficulty to user expertise.

Chapter 5

Future Work

The current work demonstrates two viable mechanisms for user-adaptive biomedical explanation generation: ontology-grounded constrained decoding and steering-vector-based activation control. While both approaches show promising results, several natural extensions remain for improving scale, grounding, and generality.

5.1 Future Directions

1. Larger-Scale Evaluation

The current constrained decoding and steering-vector experiments are conducted on a small biomedical sample set. A first priority is to scale evaluation to larger subsets of the Medical Abstracts Text Classification Dataset and, eventually, to additional biomedical NLP tasks. This would allow stronger statistical validation of readability improvements, robustness across inputs, and consistency of user-adaptive behaviour.

2. In-Domain Steering Vectors

The present steering vector is learned from contrastive explanation pairs derived from Reddit ELI5 and only later evaluated on biomedical text. A more principled next step is to construct contrastive easy-hard explanation pairs directly within the biomedical domain. In-domain steering vectors may better preserve semantic fidelity while still controlling technical complexity.

3. Hybrid Control Mechanisms

In the current report, constrained decoding and steering vectors are treated as separate control strategies. Future work could combine them into a hybrid framework in which ontology grounding provides semantic structure, constrained decoding regulates vocabulary-level readability, and steering vectors provide smoother activation-level style control. Such a system may offer both transparency and flexibility.

4. Dynamic Ontology Selection

Ontology use is currently task-dependent and largely manual. A more scalable system would dynamically choose the most relevant ontology, or combination of ontologies, for each task and input instance. This could be achieved using ontology-ranking methods or BioPortal-based retrieval, making the framework more adaptable across heterogeneous biomedical domains.

5. Human-Centred Evaluation

The current evaluation focuses mainly on readability and faithfulness metrics. Future work should incorporate human evaluation with both lay users and biomedical experts in order to assess whether the generated explanations are actually perceived as understandable, trustworthy, and useful. Such studies would provide a stronger validation of the user-adaptive goals of the system.

6. Unified Explanation System

The long-term goal is to move from task-specific explanation pipelines toward a unified biomedical explanation engine. Such a system would accept outputs from multiple biomedical predictors, ground them in suitable ontologies, and generate explanations calibrated to different expertise levels through a shared interface. The present work provides the foundation for such a system by demonstrating that both explicit decoding control and latent activation steering can serve as viable mechanisms for explanation adaptation.

Summary

Overall, future work will focus on scaling evaluation, improving domain-specific controllability, strengthening grounding, and moving toward a unified user-adaptive biomedical explanation framework.

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